

ESP32 – HC-SR04 Ultrasonic Sensor Connection Table

This document describes the physical wiring connections used in the ESP32-based IoT system. The table below clearly shows how the HC-SR04 ultrasonic sensor is connected to the ESP32 development board for distance measurement.

HC-SR04 Pin	ESP32 Pin	Connection Type	Description
VCC	5V	Power	Provides power to ultrasonic sensor
GND	GND	Ground	Common ground between ESP32 and sensor
TRIG	GPIO 5	Output	ESP32 triggers ultrasonic pulse
ECHO	GPIO 18	Input	ESP32 receives echo signal

Important Notes: • The ESP32 operates at 3.3V logic levels. • The HC-SR04 ECHO pin outputs 5V; a voltage divider is recommended to protect the ESP32 GPIO. • Female-to-male jumper wires were used due to the ESP32 board form factor. • All connections were verified before powering the board.